Notation - Dammit

Artin - linearly independence of characteristic theorem.

Def. Let G be a group and L be a field.
A group-Homomorphism $\chi : G \to L^\times$ is called a character of G in L.
The characters $\chi_1, \cdots, \chi_n$ are said to be linearly independent over L if
$\sum_{i=1}^{n} a_i \cdot \chi_i \equiv 0$ for any $a_i \in L$ implies $a_1 = \cdots = a_n = 0$.

Thm. If $\chi_1, \cdots, \chi_n : G \to L^\times$ are distinct characters of G in L,
then they are linearly independent over L.

Proof. Suppose that $\chi_1, \cdots, \chi_n$ are linearly dependent.
Choose $I \subseteq [n]$ with minimum cardinality among all non-empty sets satisfying
$\sum_{k \in I} a_k \cdot \chi_k \equiv 0$ and $\forall k \in I,\ a_k \neq 0$. $\cdots$ (*), with the coefficients a_i.
Since the characters are dependent, there are subsets of $[n]$ satisfying such conditions, so we can take this I. Moreover, since each Charac-ters have only non-zero valued, so I must have at least two elements.
Let $i, j \in I$ with $i \neq j$. Since $\chi_i \neq \chi_j$, for some $g_0 \in G$, we have

$$\chi_i(g_0) \neq \chi_j(g_0)$$

Given $g \in G$, $0 = \sum_{k \in I} a_k \cdot \chi_k(g_0 g) = \sum_{k \in I} a_k \cdot \chi_k(g_0) \cdot \chi_k(g)$. And, multiplying $\chi_j(g_0)$ to (*), $0 = \sum_{k \in I} a_k \cdot \chi_j(g_0) \cdot \chi_k(g)$. Now, substracting the two equations,
then we have $\sum_{k \in I} a_k \cdot (\chi_k(g_0) - \chi_j(g_0)) \cdot \chi_k(g) = 0$.
But, the coefficient of $\chi_j(g)$ is zero and of $\chi_i(g)$ is not zero,
so it is also non-trivial representation with the non-empty index set $J \subsetneq I$,
it contradicts to (*). Consequently, the chacters are independent. □

Note: $\mathcal{P} := \left\{ (a_1, \cdots, a_n) \in L^n \setminus \{(0, \cdots, 0)\} \,\middle|\, \sum_{k \in [n]} a_k \cdot \chi_k \equiv 0 \right\}$ is non-empty iff
$\{\chi_1, \cdots, \chi_n\}$ is linearly dependent.

View of Representation theory

Let $F^G := \{f \mid f: G \to F\}$ where G is a group and F is a field.

Then, F^G is a vector space under the operations

$$(f_1 + f_2)(g) := f_1(g) + f_2(g) \quad \text{and} \quad (af)(g) := a \cdot f(g).$$

Let $h \in G$ be given. Define

$$T_h : F^G \to F^G : f \mapsto T_h f \quad \text{where} \quad T_h f : G \to F : g \mapsto f(hg).$$

Artin's Theorem.

Lemma. Let F be a field, and $S \subseteq Aut(F)$ be a nonempty subset. Then,

$$F^S := \bigcap_{\sigma \in S} \{a \in F \mid \sigma(a) = a\}$$ is a subfield of F.

Proof. Since nontrivial intersection of fields is a field, so enough to show for any $\sigma \in S$, $\{a \in F \mid \sigma(a) = a\}$ is a field containing 0 and 1.
Given $a, b \in F$ satisfying $\sigma(a) = a$ and $\sigma(b) = b$,
$\sigma(a-b) = \sigma(a) - \sigma(b) = a - b$ and
if $b \neq 0$, then $\sigma(a \cdot b^{-1}) = \sigma(a) \cdot \sigma(b)^{-1} = a \cdot b^{-1}$, and $\sigma(0) = 0$, $\sigma(1) = 1$ clear. ∎

Thm. Let F be a field, and $G = \{\sigma_1, \cdots, \sigma_n\} \leq Aut(F)$ be a subgroup. Then, $[F : F^G] = |G|$.

More specifically, $[F : F^G] \geq |G|$ only need $G \subseteq Aut(F)$ is a subset.
Proof. Let $G \subseteq Aut(F)$ be a non-empty subset.
First, to show $[F : F^G] \geq |G|$, assume $m = [F : F^G] < |G| = n$.
Let $\{w_1, \cdots, w_m\}$ be a basis for F over F^G. Then, the system

$$\left(\sigma_j(w_i)\right)_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

has a non-trivial solution $(\beta_1, \cdots, \beta_n) \in F^n$. That is,

for all $1 \leq i \leq m$, $\sum_{k=1}^{n} \sigma_k(w_i) \cdot \beta_k = 0$.

Let $a_1, \cdots, a_m$ be arbitrary elements of F^G. Then,

for each $1 \leq i \leq m$, $0 = a_i \cdot \sum_{k=1}^{n} \sigma_k(w_i)\beta_k = \sum_{k=1}^{n} a_i \cdot \sigma_k(w_i) \cdot \beta_k = \sum_{k=1}^{n} \sigma_k(a_i \cdot w_i) \cdot \beta_k$.

Therefore,

$$0 = \sum_{i=1}^{m} \sum_{k=1}^{n} \sigma_k(a_i w_i) \cdot \beta_k = \sum_{k=1}^{n} \sum_{i=1}^{m} \sigma_k(a_i w_i) \cdot \beta_k = \sum_{k=1}^{n} \beta_k \cdot \sigma_k\left(\sum_{i=1}^{m} a_i w_i\right)$$

Since $a_1, \cdots, a_m$ was arbitrary and $w_1, \cdots, w_n$ forms a basis, thus

for any $\alpha \in F$, $\sum_{k=1}^{n} \beta_k \cdot \sigma_k(\alpha) = 0$.

But, the $\beta_1, \cdots, \beta_n$ are not all zero, so it contradicts to the independence of characters, so $[F : F^G] \geq |G|$.

[Proof of $[F:F^G] \le |G|$.

Now, assume G is a subgroup with $\sigma_1 = id$. Suppose that $[F:F^G] > |G|$. Then, there are more than $n = |G|$ numbers F^G-linearly independent elements of F, say $\alpha_1, \cdots, \alpha_{n+1}$. The system

$$\left(\sigma_i(\alpha_j)\right)_{\substack{1 \le i \le n \\ 1 \le j \le n+1}} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_{n+1} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \quad \cdots (*)$$

has a nontrivial solution $(\beta_1, \cdots, \beta_n, \beta_{n+1}) \in F^{n+1}$. Since $\sigma_1 = id$, $\alpha_1\beta_1 + \cdots + \alpha_{n+1}\cdot\beta_{n+1} = 0$ implies that at least one of $\beta_i \notin F^G$, because if all β_i are contained in F^G, then

$$\beta_1 \cdot \alpha_1 + \cdots + \beta_{n+1} \cdot \alpha_{n+1} = 0$$

is nontrivial F^G-relation of zero, so it contradicts to the independence of $\alpha_1, \cdots, \alpha_{n+1}$.

Now, let $r \in \mathbb{N}$ be the minimal number such that $(\beta_1, \cdots, \beta_r, 0, \cdots, 0) \in F^{n+1}$ satisfying $(*)$ where $\beta_1, \cdots, \beta_r$ are nonzero.

Since $\beta_1, \cdots, \beta_r$ are nonzero, dividing the equation $(*)$ as β_r, therefore we may assume $\beta_r = 1$, and since at least one of $\beta_i \notin F^G$, say β_1. Thus, for each $1 \le i \le n$, $\sigma_i(\alpha_1)\cdot\beta_1 + \cdots + \sigma_i(\alpha_{r-1})\cdot\beta_{r-1} + \sigma_i(\alpha_r) = 0$.

Since $\beta_1 \notin F^G$, so for some $k_0 \in \{1, \cdots, n\}$, $\sigma_{k_0}(\beta_1) \neq \beta_1$. For each $1 \le i \le n$,

$$\begin{aligned} 0 &= \sigma_{k_0}\left(\sigma_i(\alpha_1)\cdot\beta_1 + \cdots + \sigma_i(\alpha_{r-1})\cdot\beta_{r-1} + \sigma_i(\alpha_r)\right) \\ &= (\sigma_{k_0}\circ\sigma_i)(\alpha_1)\cdot\sigma_{k_0}(\beta_1) + \cdots + (\sigma_{k_0}\circ\sigma_i)(\alpha_{r-1})\cdot\sigma_{k_0}(\beta_{r-1}) + (\sigma_{k_0}\circ\sigma_i)(\alpha_r) \end{aligned}$$

Since G is a group, so $\sigma_{k_0}G = G$, therefore the equation above means

$$= \sigma_j(\alpha_1)\cdot\sigma_{k_0}(\beta_1) + \cdots + \sigma_{k_0}(\alpha_{r-1})\cdot\sigma_{k_0}(\beta_{r-1}) + \sigma_j(\alpha_r) \quad \text{where } \sigma_j = \sigma_{k_0}\circ\sigma_i.$$

Now, for $i = 1, \cdots, n$, $\sigma_i(\alpha_1)\cdot(\beta_1 - \sigma_{k_0}(\beta_1)) + \cdots + \sigma_i(\alpha_{r-1})\cdot(\beta_{r-1} - \sigma_{k_0}(\beta_{r-1})) = 0$, and $\beta_1 - \sigma_{k_0}(\beta_1) \neq 0$, so it satisfies $(*)$ with nontrivial solution, it contradicts to the minimality. ∎

Corollaries.

Corollary. Let K/F be a finite extension. Then

$$|Aut(K/F)| \leq [K:F]$$

and the equality holds if and only if $F = K^{Aut(K/F)}$.

Proof. Let $F_1 = K^{Aut(K/F)}$. Then, $F \subseteq F_1 \subseteq K$. By the Artin's theorem, $[K:F_1] = |Aut(K/F)|$, hence $[K:F] = [K:F_1] \cdot [F_1:F] = |Aut(K/F)| \cdot [F_1:F]$. ☐

Corollary. Let F be a field and $G \leq Aut(F)$ be a finite subgroup. Then

F/F^G is always a Galois extension with $Gal(F/F^G) = G$.

Proof. By definition, $G \leq Aut(F/F^G)$. So, $|G| \leq |Aut(F/F^G)|$.
And, by the Artin's theorem, $|G| = [F:F^G]$ and now by the previous corollo-rary, we obtain $[F:F^G] = |G| \leq |Aut(F/F^G)| \leq [F:F^G]$, So proves. ☐

Corollary. Let F be a field, and $G_1, G_2 \leq Aut(F)$ be subgroups. Then,

$G_1 = G_2$ if and only if $F^{G_1} = F^{G_2}$.

Proof. Suppose that $F^{G_1} = F^{G_2}$. Then, $Aut(F/F^{G_1}) \subseteq G_2$, by definition, and the previous lemma gives that

$$G_1 = Aut(F/F^{G_1}) \subseteq G_2.$$

Similarly $G_2 \subseteq G_1$, so $G_1 = G_2$. ☐

Normal basis theorem.

Def. Let E/F be a Galois extension field of degree of n.
For fixed $\gamma \in E$, a set

$$\eta = \{\sigma_1(\gamma), \cdots, \sigma_n(\gamma)\} \quad \text{where } Gal(E/F) = \{\sigma_1, \cdots, \sigma_n\}.$$

is called a normal basis for E/F if η is a basis.

Lemma. Let E/F be Galois extension with $Gal(E/F) = \{\sigma_1, \cdots, \sigma_n\}$.
For a basis $\beta = \{e_1, \cdots, e_n\}$ for E over F,

$$\mathcal{A} = \{(\sigma_1(e_i), \cdots, \sigma_n(e_i)) \mid 1 \le i \le n\}$$

forms an E-basis for E^n

Proof. Let $W = Span_L(\mathcal{A}) \le E^n$. Suppose that $W \neq E^n$.
Then, there exists a nonzero linear functional $\varphi : E^n \to E$ which vanishes on W. (In other words, $W \neq E^n$ implies $W^\circ \neq (E^n)^\circ = \{0\}$.)
Let $\{f_1, \cdots, f_n\}$ be the dual basis for $(E^n)^*$ corresponding to the standard basis $\{\alpha_1, \cdots, \alpha_n\}$ for E^n. For each $1 \le i \le n$,

$$\varphi\left((\sigma_j(e_i))_{j=1}^n\right) = \sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j(e_i) = 0, \text{ since } \varphi \in W^\circ.$$

For any $x \in E$, x is F-linear combination by β, so that

$$x = c_1 e_1 + \cdots + c_n e_n \text{ for some } c_i \in F$$

Now, Since each σ_i's are field automorphism which F fixes, so

$$\begin{aligned}\sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j(x) &= \sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j\left(\sum_{i=1}^n c_i e_i\right) \\ &= \sum_{j=1}^n \varphi(\alpha_j) \cdot \sum_{i=1}^n c_i \cdot \sigma_j(e_i) \\ &= \sum_{i=1}^n c_i \cdot \sum_{j=1}^n \varphi(\alpha_j) \cdot \sigma_j(e_i) = \sum_{i=1}^n c_i \cdot 0 = 0.\end{aligned}$$

Now, by the independentness of the characters $\sigma_1|_{E^\times}, \cdots, \sigma_n|_{E^\times}$,

$$\varphi(\alpha_1) = \cdots = \varphi(\alpha_n) = 0.$$

But, it implies $\varphi \equiv 0$, so contradiction. □

Thm. Every finite Galois extension of an infinite field has a normal basis

Proof. Let E/F be the finite Galois extension. Denote

$$Gal(E/F) = \{\sigma_1, \cdots, \sigma_n\} \text{ where } n = [E:F], \text{ and } \sigma_1 = id.$$

Given $x \in E$, consider F-linear relation

$$\sum_{j=1}^{n} a_j \cdot \sigma_j(x) = 0 \quad \text{for some } a_1, \cdots, a_n \in F.$$

For each $1 \le i \le n$,

$$\sigma_i^{-1}\left(\sum_{j=1}^{n} a_j \cdot \sigma_j(x)\right) = \sum_{j=1}^{n} a_j \cdot (\sigma_i^{-1} \circ \sigma_j)(x) = \sigma_i^{-1}(0) = 0, \text{ so that}$$

$$\begin{pmatrix} (\sigma_1^{-1}\circ\sigma_1)(x) & \cdots & (\sigma_1^{-1}\circ\sigma_n)(x) \\ \vdots & \ddots & \\ (\sigma_n^{-1}\circ\sigma_1)(x) & \cdots & (\sigma_n^{-1}\circ\sigma_n)(x) \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \cdots (*)$$

Therefore, the linear relation above is trivial if and only if the equation $(*)$ has only trivial solution if and only if the matrix in $(*)$ has nonzero determinant.

Let $\beta = \{\alpha_1, \cdots, \alpha_n\}$ be a basis for E over F.

And, write an arbitrary $x \in E$ as $x = \sum_{k=1}^{n} a_k \cdot \alpha_k$ where $a_k \in F$. Then, for each $1 \le i, j \le n$, $(\sigma_i^{-1} \circ \sigma_j)(x) = \sum_{k=1}^{n} a_k \cdot (\sigma_i^{-1} \circ \sigma_j)(\alpha_k)$.

Consider the Polynomial on variables $X_1, \cdots, X_n$ over E:

$$\Delta(X_1, \cdots, X_n) \overset{def}{=} \det\left(\sum_{k=1}^{n} (\sigma_i^{-1} \circ \sigma_j)(\alpha_k) \cdot X_k\right)_{\substack{1 \le i \le n \\ 1 \le j \le n}}$$

Proof, Page 2.

Meanwhile, observe that: Consider a matrix

$$\left(\sigma_i(\alpha_j)\right)_{i,j} \in M_{n\times n}(E)$$

Then, it is invertible, because: assume that this matrix is not invertible. Then, the rows are linearly dependent, so that there are not all zero scalars $a_1, \cdots, a_n \in E$ s.t

$$\text{for all } 1\le k\le n,\ \sum_{i=1}^{n} a_i \cdot \sigma_i(\alpha_k) = 0.$$

Now, let $x \in E$ be given. Denote $x = \sum_{k=1}^{n} b_k \alpha_k$ where $b_k \in F$.

Then, $\sum_{i=1}^{n} a_i \sigma_i(x) = \sum_{i=1}^{n} a_i \cdot \sigma_i\left(\sum_{k=1}^{n} b_k \cdot \alpha_k\right) = \sum_{k=1}^{n} b_k \cdot \sum_{i=1}^{n} a_i \cdot \sigma_i(\alpha_k) = 0.$

But, it contradicts to the independence of characters.

Now, the equation

$$\begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_1(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_n(\alpha_1) & \cdots & \sigma_n(\alpha_n) \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

has a unique solution, $(c_1, \cdots, c_n) \in E^n$. Since $\sigma_1 = id$, so that

$$\sum_{k=1}^{n} c_k \cdot \alpha_k = 1 \quad \text{and} \quad \sum_{k=1}^{n} c_k \cdot \sigma_j(\alpha_k) \quad (2 \le j \le n).$$

That is, $\sum_{k=1}^{n} c_k \cdot (\sigma_i^{-1} \circ \sigma_j)(\alpha_k) = \delta_{i,j}$. Therefore, $\Delta(c_1, \cdots, c_n) = \det I_n = 1$. So Δ is non-zero polynomial.

So, the non-zero polynomial Δ has only finite number roots and since F is infinite, there are $b_1, \cdots, b_n \in F$ such that $\Delta(b_1, \cdots, b_n) \neq 0$.

Now, $0 \neq \Delta(b_1, \cdots, b_n) = \det\left(\sum_{k=1}^{n} (\sigma_i^{-1} \circ \sigma_j)(\alpha_k) \cdot b_k\right) = \det\left((\sigma_i^{-1} \circ \sigma_j)(x)\right)$

where $x = \sum_{k=1}^{n} b_k \cdot \alpha_k$, so

$$\{\sigma_1(x), \cdots, \sigma_n(x)\} \text{ is a normal basis for } E \text{ over } F. \square$$

Thm. Every finite cyclic extension E/F has a normal basis.
Proof.

Thm. Let E/F be a Galois extension with a normal basis $\{\sigma(\alpha) \mid \sigma \in Gal(E/F)\}$.

For any subgroup $H < Gal(E/F)$, $E^H = F(\alpha_H)$ where $\alpha_H = Tr_{E/E^H}(\alpha)$.

In Particular, if $N \triangleleft Gal(E/F)$, then a normal basis for E^N over F is $\{\tau(\alpha_N) \mid \tau \in Gal(E/F)/N\}$ where $\alpha_N = Tr_{E/E^N}(\alpha)$.